NAVEEN NALLAMOTHU

AI ENGINEER | AZURE AI ENGINEER ASSOCIATE

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PROFESSINAL SUMMARY

- AI Engineer with 4+ years of experience across machine learning and AI engineering, including designing and building production LLM systems on Microsoft Azure.
- Designed and built an end-to-end, evidence-backed due-diligence platform that extracts ~350 claims per case from unstructured company documents and verifies each against evidence.
- Hands-on with Azure OpenAI and Microsoft Foundry for LLM-based structured extraction, reaching 92% claim-extraction accuracy on a 1,200-claim labelled evaluation set.
- Experienced in agentic AI: built a seven-agent verification system on Foundry Agent Service with a deterministic orchestrator and tools exposed through Model Context Protocol (MCP).
- Strong in retrieval: implemented hybrid RAG on Azure AI Search (vector, keyword and semantic ranking) and GraphRAG, improving exact-identifier retrieval by 28%.
- Integrated eight Equifax APIs as middleware with Pydantic v2 schemas and idempotent request handling, governed through Azure API Management.
- Security-focused: credentials managed in Azure Key Vault and accessed through Managed Identity, per-case access control with Microsoft Entra ID, agent guardrails and auditable human-in-the-loop review.
- Cost and performance aware: end-to-end tracing with Application Insights, per-case token and cost tracking, and model tiering that cut cost per case by 38%.
- Solid machine learning foundation in classification, regression and clustering, with Python data pipelines and REST API integration.
- Microsoft Certified: Azure AI Engineer Associate (AI-102) and Azure Data Scientist Associate (DP-100); MS in Data Science.

TECHNICAL SKILLS

Programming Languages	Python, SQL, Bash
Generative AI & LLMs	Azure OpenAI, Microsoft Foundry, GPT Models, Llama, Hugging Face, Prompt Engineering, Context Engineering, Structured Outputs (JSON Schema), Function/Tool Calling, Model Tiering, Hallucination Control
Agentic AI	Foundry Agent Service, Multi-Agent Orchestration, Model Context Protocol (MCP), LangChain, LangGraph, Semantic Kernel, Agent Memory and State, Human-in-the-Loop Workflows
RAG & Retrieval	Azure AI Search, Hybrid Retrieval (Vector, Keyword, Semantic Ranking), Embeddings, Chunking Strategies, Re-Ranking, GraphRAG, Knowledge Graphs, Entity Resolution, pgvector, FAISS, Pinecone
Document AI	Azure AI Document Intelligence, Content Understanding, OCR, Table and Layout Extraction
Machine Learning	Scikit-learn, XGBoost, PyTorch, TensorFlow, Classification, Regression, Clustering, Decision Trees, KNN, SVM, NLP, Feature Engineering, Model Evaluation, Hyperparameter Tuning
Data Science	Pandas, NumPy, Statistical Analysis, Hypothesis Testing, Exploratory Data Analysis, Data Preprocessing and Validation, Matplotlib, Seaborn, Power BI
Evaluation & LLMOps	LLM and RAG Evaluation (Groundedness, Relevance, Faithfulness), RAGAS, Foundry Evaluation, Regression Test Sets, OpenTelemetry, Application Insights, Azure Monitor, Token and Cost Tracking
MLOps & Deployment	Azure Machine Learning, MLflow, Docker, Kubernetes (AKS), CI/CD (GitHub Actions, Azure DevOps), Terraform, Git
Backend & APIs	FastAPI, Pydantic v2, REST APIs, Azure Functions, Azure API Management, Azure Logic Apps, Idempotent API Design
Databases & Storage	Azure Cosmos DB, Azure Blob Storage, Azure Databricks, PySpark

Security & Responsible AI	Microsoft Entra ID, Managed Identity, Azure Key Vault, Azure AI Content Safety, Guardrails, Prompt-Injection Defence, PII Detection, Role-Based Access Control, Audit Trails
Third-Party Integrations	Equifax APIs (OneView Commercial Credit Report, Credit Report v2, Business Verify, B2B Connect, Digital Identity Trust, AML Connect, Employment Select All, Talent Report Education)

CERTIFICATIONS

- **Microsoft Certified:** Azure AI Engineer Associate (AI-102)
- **Microsoft Certified:** Azure Data Scientist Associate (DP-100)

EDUCATION

Masters of Science in Data Science — Mar 2023 – May 2024
Saint Peter's University — Jersey City, NJ
Bachelors of Technology in Computer Science — Jun 2018 – Jun 2022
Jawaharlal Nehru Technological University — Hyderabad, India

EXPERIENCE

AI Engineer — **Aug 2024 – Present**
Fact Finders Pro — Remote, USA

- Designed and built an end-to-end, evidence-backed due-diligence platform on Azure that converts pitch decks, financials and cap tables into atomic, independently verified claims, processing ~25 documents and ~350 claims per case.
- Built the ingestion pipeline with Azure Blob Storage and Azure AI Document Intelligence to extract text, tables and layout with page-level provenance, so every finding traces to its source document and page.
- Developed LLM-based claim extraction on Azure OpenAI (Microsoft Foundry) that returns typed, structured JSON claims across financial, IP, market, team and legal categories, achieving ~92% extraction accuracy on a 1,200-claim labelled evaluation set.
- Architected a multi-stage, multi-agent verification system on Foundry Agent Service with seven domain agents (team, financial, market, competition, IP, technology, legal) behind a deterministic orchestrator, keeping LLM calls out of routing to cut cost and latency.
- Implemented hybrid RAG on Azure AI Search (vector, keyword and semantic ranking) with chunk-level source metadata, improving retrieval of exact identifiers such as patent numbers by ~28% over vector-only search.
- Built agent tools as Azure Functions exposed through MCP and governed by Azure API Management (authentication, rate limiting, quotas, logging), giving agents controlled access to Equifax and five other external data providers.
- Integrated eight Equifax APIs as a middleware layer with Pydantic v2 schemas and idempotent request handling: commercial credit (OneView Commercial Credit Report, Credit Report v2), business verification (Business Verify, B2B Connect), identity and compliance (Digital Identity Trust, AML Connect), and background checks (Employment Select All, Talent Report Education).
- Secured all Equifax credentials (client ID, client secret, member ID and member numbers) in Azure Key Vault, retrieved at run time through Managed Identity with no secrets in code or configuration.
- Designed the claim and evidence data model in Azure Cosmos DB, plus a rule-and-LLM verification engine that classifies each claim as verified, partially verified, conflict or unverified, with confidence and severity scores.
- Developed a hybrid scoring engine that combines deterministic calculations with LLM semantic reasoning to evaluate founder profiles and pitch decks.
- Designed structured prompt engineering workflows that reduce hallucinations and keep model outputs consistent across runs.

- Built an entity graph (GraphRAG) linking people, companies, patents and investors to surface multi-hop relationships that document retrieval alone misses.
- Implemented a human-in-the-loop review workflow that routes high-severity conflicts to analysts with full evidence and stores each decision as an auditable event.
- Enforced per-case access control with Microsoft Entra ID and added agent guardrails that report evidence conflicts without asserting intent.
- Created a stateless dual-interface service with FastAPI, giving engineering and product teams a shared way to call the pipeline.
- Instrumented end-to-end tracing with Application Insights and per-case token and cost tracking, and applied model tiering to reduce cost per case by ~38% and processing time to ~12 minutes.

Environment: Python, FastAPI, Pydantic v2, Microsoft Foundry, Azure OpenAI, Foundry Agent Service, Azure AI Search, Azure AI Document Intelligence, Content Understanding, Azure Blob Storage, Azure Cosmos DB, Azure Functions, Azure Logic Apps, Azure API Management, Model Context Protocol (MCP), RAG, GraphRAG, Microsoft Entra ID, Managed Identity, Azure Key Vault, Azure AI Content Safety, Application Insights, Azure Monitor, Equifax APIs (OneView Commercial Credit Report, Credit Report v2, Business Verify, B2B Connect, Digital Identity Trust, AML Connect, Employment Select All, Talent Report Education), REST APIs, JSON, Git

Machine Learning Engineer **May 2021 – Jan 2023**
Tech Mahindra Hyderabad, India

- Built machine learning models for enterprise use cases, including Decision Trees, KNN, SVM, Linear Regression and clustering algorithms.
- Evaluated and compared model performance across algorithms to select the best fit for each use case.
- Automated data preprocessing, cleaning and validation pipelines in Python using object-oriented design.
- Integrated REST APIs to connect internal systems with third-party services.
- Delivered work in Agile sprint cycles and maintained code through Git-based version control workflows.

Environment: Python, Scikit-learn, Pandas, NumPy, Decision Trees, KNN, SVM, Linear Regression, Clustering, REST APIs, Git, Agile